

19/4/2026

BITI 2213: Knowledge Based System

Logic:

1. Propositional Logic
2. Predicate Logic

Introduction to Logic

- ❑ Logic is a form of knowledge representation.
- ❑ Two kinds of logic: propositional logic and predicate logic.

Propositional and Predicate Logic

Propositional Logic

- The study of **statements** and their **connectivity** structure.

Predicate Logic

- The study of **individuals** and their **properties**.

Propositional logic more abstract and hence **less detailed** than predicate logic.

PROPOSITIONAL LOGIC

Propositional Logic

- ❑ The simplest form of logic.
- ❑ Assists human in reasoning by providing a mechanism that enables us to first **evaluate simple statements** and next, **complex statements** formed through the use of connectives.
- ❑ The connective symbols: **$\wedge$** (AND), **$\vee$** (OR), **$\sim$** (NOT), **$\rightarrow$** (IMPLICATION), **$\equiv$** (EQUIVALENCE).

Propositional Logic

- ❑ A proposition is a **simple statement** having a value of either true (**T**) or false (**F**).
- ❑ Simple statements are typically used to express some **states or facts** about the world.
- ❑ For example:
 - Today is Monday
 - It rained yesterday
 - It is cold
 - It is 2 pm
 - We have an assignment

Propositional Logic

- ❑ Propositional logic allows us to assign a truth table to each of the entire statements.
- ❑ But has no facilities to analyze the individual words that compose the statements.
- ❑ So, the representation of the simple statements (in the previous slide) is

today-is-monday
it-rained-yesterday
it-is-cold
it-is-2-pm
we-have-an-assignment

Propositional Logic

- ❑ Propositions can also be combined with other propositions in order to express more complex concepts about the world.

- ❑ For example:
(Today is Monday) and (It is 2 pm)

can be represented in propositional logic as
today-is-monday and it-is-2 pm

- ❑ Can be assigned into symbols such as
P = today-is-monday
Q = it-is-2-pm

Propositional Logic

The Fundamental Unit of Logic: The Proposition

- Notion of “truth” and “falsity” from our everyday conversation.
- A **proposition** is a statement that is **either true or false** (not both, and not neither) at some given time.
 - “I am a cup of tea” is a proposition.
 - “I hate tea” is a proposition.
 - “Drink this tea” is NOT a proposition.

Propositional Logic

The Binary Connectives

- More interesting statements can be created through the combination of two propositions using **binary connectives**.
- A binary connective is a method for determining the aggregate truth value of two propositions – the result of connecting two propositions is another proposition.
- Common binary connectives are AND, OR, and XOR, and they get their meanings from English.

Propositional Logic

Meanings of the Connectives

AND ($\wedge$)

- If p is a proposition, and q is a proposition, then $p \wedge q$ is true exactly when both p and q are true ("I hate coffee AND I hate tea").

OR ($\vee$)

- $p \vee q$ is true if either p is true or q is true ("She can visit on Sunday OR Tuesday") -- both can be true.

XOR ($\oplus$)

- $p \oplus q$ is true if either p is true or q is true but NOT both ("You can have soup OR salad").

English is a terribly ambiguous language! (when it comes to logic).

Propositional Logic

Implications

- Another common form of binary connective is the **implication** ($\rightarrow$)
 - Stated as "if p , then q "; written $p \rightarrow q$
 - Alternately: " p implies q ." " q if p ."
 - "If it is raining, then we won't go to the beach."
- ***Implications are one-way!***
 - Nothing says we *will* go to the beach if it's not raining, but we definitely *won't* go if it is.

Propositional Logic

Negations

- If p is a proposition, then $\neg p$ is its **negation** (opposite)
 - p = "I like tea.",
 - $\neg p$ = "I don't like tea."
 - p = "Bob's new car is red.",
 - $\neg p$ = "Bob's new car is not red."
 - $\neg p$ is true exactly when p is false, and vice versa.
- Negation is a unary operator on propositions
-precedence over binary.

Propositional Logic

A Propositional Calculus

- Represent absolute truth with **T**, absolute falsity with **F**.
- Assign letters to various propositions.
- Binary connectives are used to join the propositions.
- GOAL: Particular assignments of **T** and **F** to each proposition consistently yield the overall truth value of the proposition as a whole.

Propositional Logic

Truth Tables

p	q	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \rightarrow q$
T	T	T	T	F	T
T	F	F	T	T	F
F	T	F	T	T	T
F	F	F	F	F	T

Propositional Logic

Logical Equivalence

- Two compound propositions over some set of component propositions are **logically equivalent** (denoted $\Leftrightarrow$), if they have the same truth value for each possible assignment to components.
 - This also means that they are logically equivalent if they
imply each other, i.e. if $(p \rightarrow q) \wedge (q \rightarrow p)$ is always true.
- Can be written as “ p iff q ” (p if and only if q)
- Leads to some well-known reductions.

Propositional Logic

Common Equivalences

- Used to reduce complex propositions
 - De Morgan 1: $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$
 - De Morgan 2: $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$
 - Implication: $p \rightarrow q \Leftrightarrow \neg p \vee q$
 - Double negation: $\neg(\neg p) \Leftrightarrow p$
 - Distributive 1: $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$
 - Distributive 2: $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$
- Replace parts of complex propositions with logically equivalent ones.

Propositional Logic

- Propositional logic provides a mechanism for assigning a truth value to compound propositions based upon the truth values of the individual simple propositions and the nature of the connective involved.

Table 2.4 Truth Table of the Binary Logical Connectives

p	q	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

Table 2.5 Truth Table of Negation Connectives

p	$\sim p$
T	F
F	T

PREDICATE LOGIC

Predicate Logic

- ❑ Propositional logic's weakness is its limited ability to express knowledge.
- ❑ In propositional logic there is no way to access the components of an individual assertion.

$P = \text{it-is-2-pm}$

- ❑ Alternatively,  Cannot change any of the components since entire sentence "it is 2 pm" is denoted by the symbol P.
it is 2 pm can be represented as a predicate as **pm(2)**

Predicate Logic

- ❑ Examples.

mortal(socrates)
colour(grass, green)
colour(sky, blue)

- ❑ The predicate can be manipulated to access their individual components and inferring new sentences.

Predicate Logic

- In predicate logic, a proposition is divided into two parts
 - ▶ the predicate (or assertion),
 - ▶ the arguments (or objects)
- Predicate logic allows to break a statement down into component parts
 - ▶ an object,
 - ▶ characteristic of an object, or
 - ▶ some assertion about an object

□ Examples.

mortal(socrates)
colour(grass, green)
colour(sky, blue)

Predicate Logic

- ❑ The predicate **can be manipulated** to access their individual components and inferring new sentences.
- ❑ The predicate allows **expressions to contain variables** – let us create general assertions about classes of entities.

Example:

pm(X)

could be read as for all values of X, where X is a time in the afternoon or evening, the statement pm(X) is true.

Predicate Logic

- ❑ The semantic of predicate calculus provide a formal basis for determining the truth value of well-formed expressions.
- ❑ The truth of expressions depends on the mapping of **constants**, **variables**, **predicates**, and **functions** into objects and relations in the domain of discourse.
- ❑ Example:

pm(2) → have(we, class)

Predicate Logic

- ❑ $\text{mortal}(X) \rightarrow \text{man}(X)$
- ❑ Quantification of variables is an important part of predicate logic semantics.
- ❑ Variables must be quantified in either; ***universally*** or ***existentially***.
- ❑ A variable is considered *free* if it is not within the scope of either the universal or existential quantifiers.
- ❑ Is *closed* if all of its variables are quantified.

Predicate Logic

- ❑ Symbol for universal quantifier is $\forall$.

Example: $\forall X, \text{boy}(X) \rightarrow \text{likes}(X, \text{girl})$

indicates that X is universally quantified in boy(X).

- ❑ Symbol for existential quantifier is $\exists$.

Example: $\exists X, \text{girl}(X) \rightarrow \text{likes}(X, \text{doll})$

indicates that some of X exist in girl(X).

Predicate Logic

- ❑ Many grammatically correct English sentences can be represented in the predicate logic using symbols, connectives, and variable symbols.
- ❑ There is no unique mapping of sentences into predicate calculus.
- ❑ Example:

If it doesn't rain on Monday, Tom will go to the mountains.

$\sim \text{weather}(\text{rain}, \text{monday}) \rightarrow \text{go}(\text{tom}, \text{mountains})$

Some people likes banana.

$\exists X (\text{people}(X) \wedge \text{likes}(X, \text{banana}))$

Exercise

$\forall x (\text{student}(x) \rightarrow \text{study}(x))$

$\forall x (\text{elephant}(x) \rightarrow \neg \text{cry}(x))$

$\exists x (\text{table}(x) \wedge \neg \text{numlegs}(x, 4))$

$\forall x (\text{malaysian}(x) \rightarrow (\text{supports}(x, \text{Man.United}) \vee \text{supports}(x, \text{Chelsea})))$

$\neg \exists x (\text{cigarette}(x) \wedge \text{good}(x))$

Answer

- ☐ All student study.
- ☐ All elephant not cry.
- ☐ There is some table that doesn't have 4legs.
- ☐ All Malaysians support either Man. United or Chelsea.
- ☐ There is no cigarette which is good.

LOGICAL REASONING

Logical Reasoning

- ❑ **Infer**: to conclude or decide from something known or assumed; derive by reasoning; draw as a conclusion.
- ❑ **Reason**: to think coherently and logically (rationally); draw inferences or conclusions from facts known or assumed.
- ❑ The process of **reasoning** involves **making inferences** from known facts.
- ❑ **Automated reasoning** is described as:
Given a set of premises known and a reasoning method, certain conclusions can be inferred to also be true.

Logical Reasoning

□ Deduction:

$\forall X, \forall Y, \forall Z, \text{larger}(X, Y) \wedge \text{larger}(Y, Z) \rightarrow \text{larger}(X, Z)$

$\text{larger}(\text{house}, \text{car})$

$\text{larger}(\text{car}, \text{cat})$

$\text{larger}(\text{cat}, \text{mouse})$

□ Induction:

$\forall X, \text{snorkeling}(X) \rightarrow \text{wet}(X)$

$\text{snorkeling}(\text{imran})$

$\text{wet}(\text{azrin})$

Premise

- ◆ Facts
- ◆ Sentences
- ◆ Observations
- ◆ Information



Logical
Reasoning



Inference

- ◆ New facts
- ◆ Actions
- ◆ Conclusions

Automated Theorem Proving and Resolution

- ❑ Resolution is a form of inferencing based on the sound rule of inference by the same name.

- ❑ Resolution Rule states:

IF $(A \vee B)$ is true and $(\sim B \vee C)$ is true,
THEN $(A \vee C)$ is true.

- ❑ The process of resolution operates by finding two clauses (e.g., $A \vee B$ and $(\sim B \vee C)$) that are *resolved* to produce a new clause (i.e., $A \vee C$) called the *resolvent*.

Automated Theorem Proving and Resolution

- ❑ For example:

$$\forall X, \text{snorkeling}(X) \rightarrow \text{wet}(X) \quad (i)$$

$$\text{snorkeling}(\text{imran}) \quad (ii)$$

- ❑ Based on The Idempotent Law,

$$\forall X, p(X) \rightarrow q(X) \equiv \sim p(X) \vee q(X)$$

- ❑ Using this law on (i)

$$\forall X, \text{snorkeling}(X) \rightarrow \text{wet}(X)$$

results in

$$\sim \text{snorkeling}(X) \vee \text{wet}(X)$$

Unifying X with *imran* and combining this with (ii), the resolution results in $\text{wet}(\text{imran})$ is derived as true.

PROS N CONS

Predicate Logic: Pros and Cons

- ❑ PROLOG is a logic programming language, is a powerful formalism for developing a knowledge-based system.
- ❑ It is an established means of representing knowledge and its built-in reasoning mechanism using deduction is theoretically founded.

Some cons:

1. Unable to manage **uncertainty** facts.
2. **Unsuitable for real-word problem** domains which involve with time.
3. Declarative versus procedural programming.

CONCLUSION

See you in the lab session!